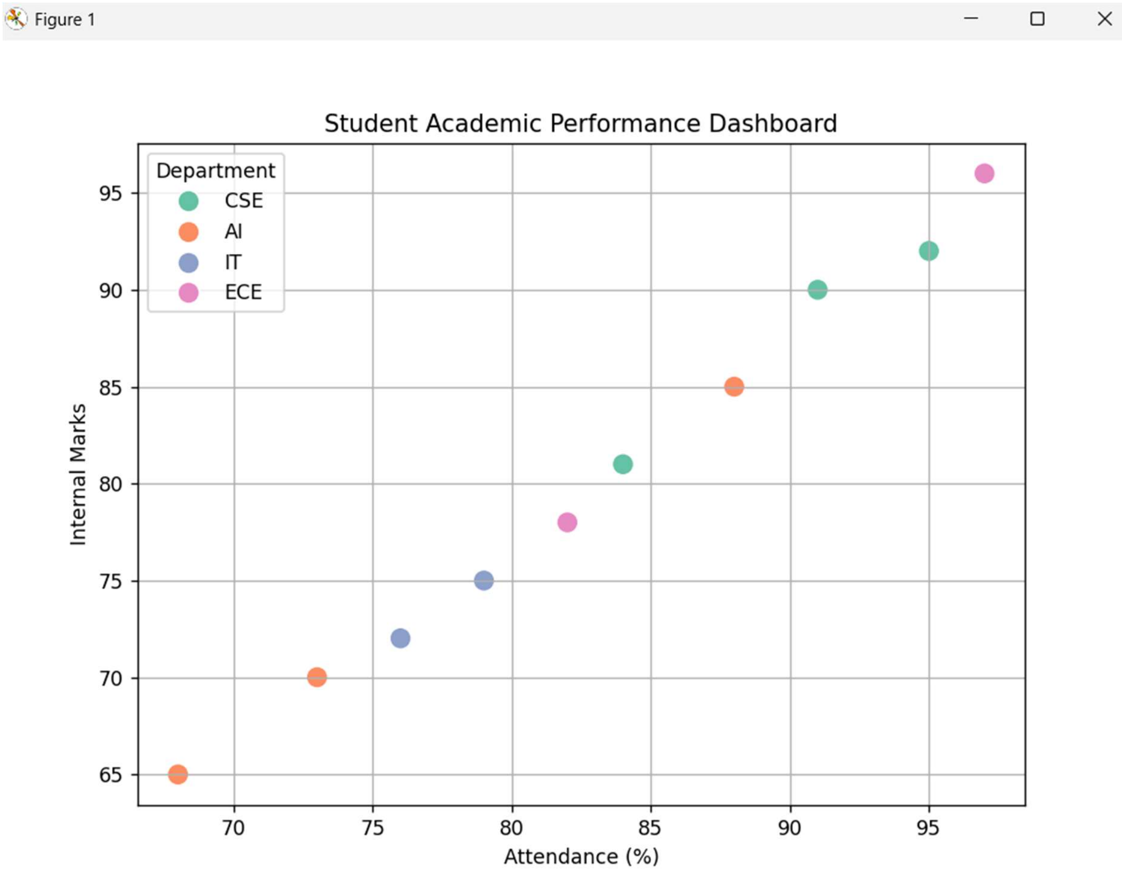
Scenario 1: Student Academic Performance Dashboard

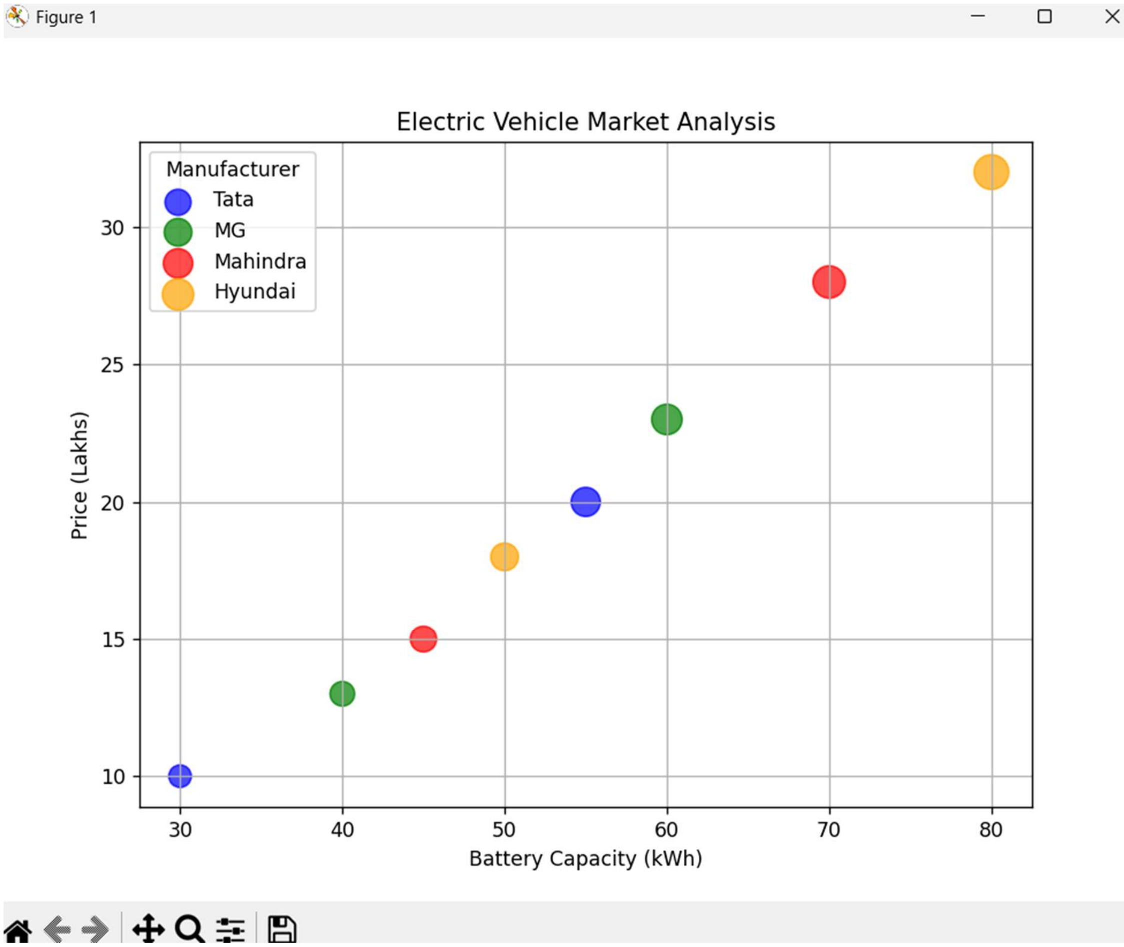
```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


data = {

    "Student ID":
["S001","S002","S003","S004","S005","S006","S007","S008","S009","S010"],

    "Attendance (%)": [95,88,76,82,91,68,79,97,84,73],

    "Internal Marks": [92,85,72,78,90,65,75,96,81,70],

    "Department": ["CSE","AI","IT","ECE","CSE","AI","IT","ECE","CSE","AI"]

}


df = pd.DataFrame(data)


plt.figure(figsize=(8,6))

sns.scatterplot(data=df,

                x="Attendance (%)",

                y="Internal Marks",

                hue="Department",

                palette="Set2",

                s=120)


plt.title("Student Academic Performance Dashboard")

plt.xlabel("Attendance (%)")

plt.ylabel("Internal Marks")

plt.legend(title="Department")

plt.grid(True)

plt.show()
```



Scenario 2: Electric Vehicle Market Analysis

```
import pandas as pd

import matplotlib.pyplot as plt


data = {

    "Vehicle Model":["EV1","EV2","EV3","EV4","EV5","EV6","EV7","EV8"],

    "Battery Capacity":[30,40,45,50,55,60,70,80],

    "Price":[10,13,15,18,20,23,28,32],

    "Driving Range":[250,300,340,380,420,460,520,600],

"Manufacturer":["Tata","MG","Mahindra","Hyundai","Tata","MG","Mahindra","Hyundai"]

}
```

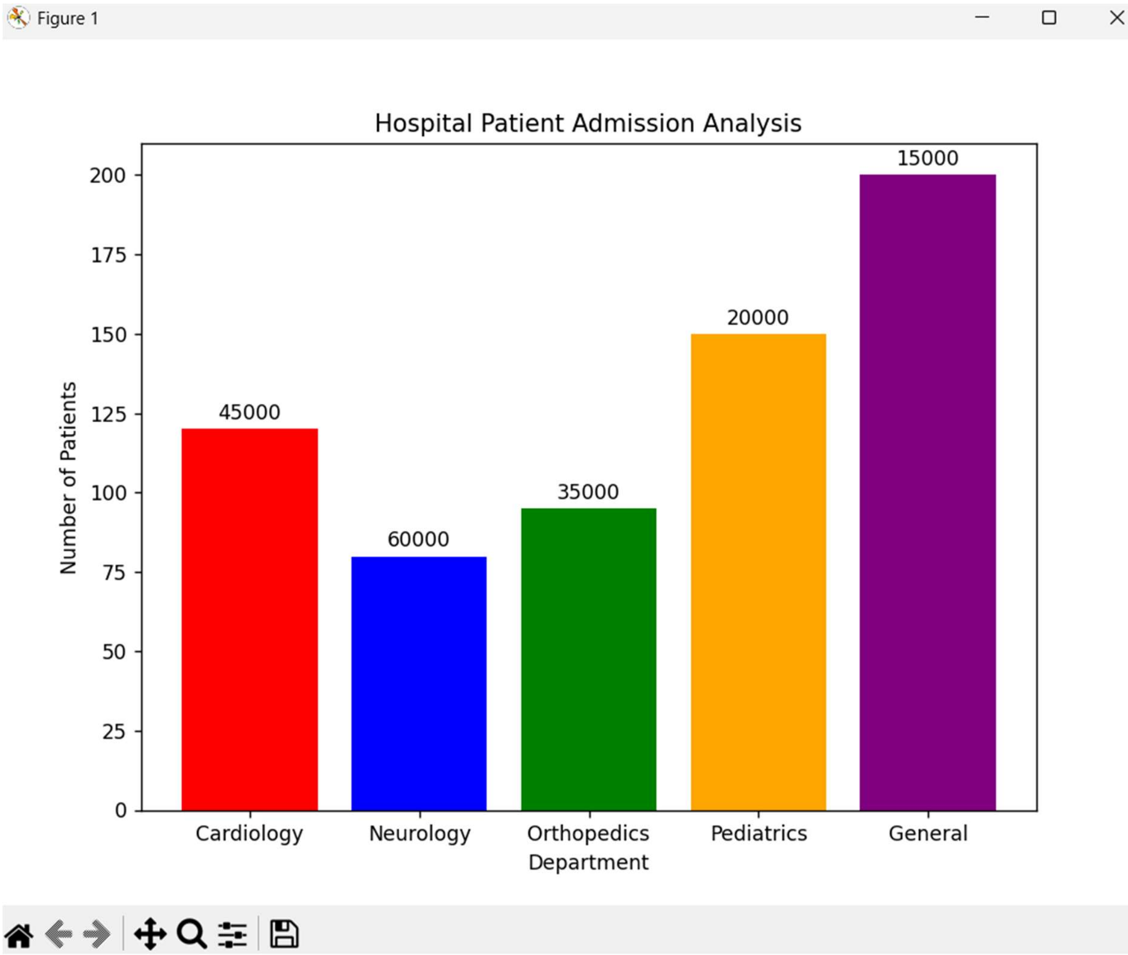
```
df=pd.DataFrame(data)

colors={"Tata":"blue","MG":"green","Mahindra":"red","Hyundai":"orange"}

plt.figure(figsize=(8,6))

for m in df["Manufacturer"].unique():
    d=df[df["Manufacturer"]==m]
    plt.scatter(d["Battery Capacity"],
                d["Price"],
                s=d["Driving Range"]/2,
                color=colors[m],
                alpha=0.7,
                label=m)

plt.title("Electric Vehicle Market Analysis")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price (Lakhs)")
plt.legend(title="Manufacturer")
plt.grid(True)
plt.show()
```



Scenario 3: Hospital Patient Admission Analysis

```
import pandas as pd

import matplotlib.pyplot as plt


data={

   "Department":["Cardiology","Neurology","Orthopedics","Pediatrics","General"],

   "Number of Patients":[120,80,95,150,200],

   "Average Treatment Cost":[45000,60000,35000,20000,15000]

}


df=pd.DataFrame(data)


colors=["red","blue","green","orange","purple"]
```

```
plt.figure(figsize=(8,6))

bars=plt.bar(df["Department"],
             df["Number of Patients"],
             color=colors)

for bar,cost in zip(bars,df["Average Treatment Cost"]):
    plt.text(bar.get_x()+bar.get_width()/2,
             bar.get_height()+3,
             str(cost),
             ha="center")

plt.title("Hospital Patient Admission Analysis")
plt.xlabel("Department")
plt.ylabel("Number of Patients")

plt.show()
```

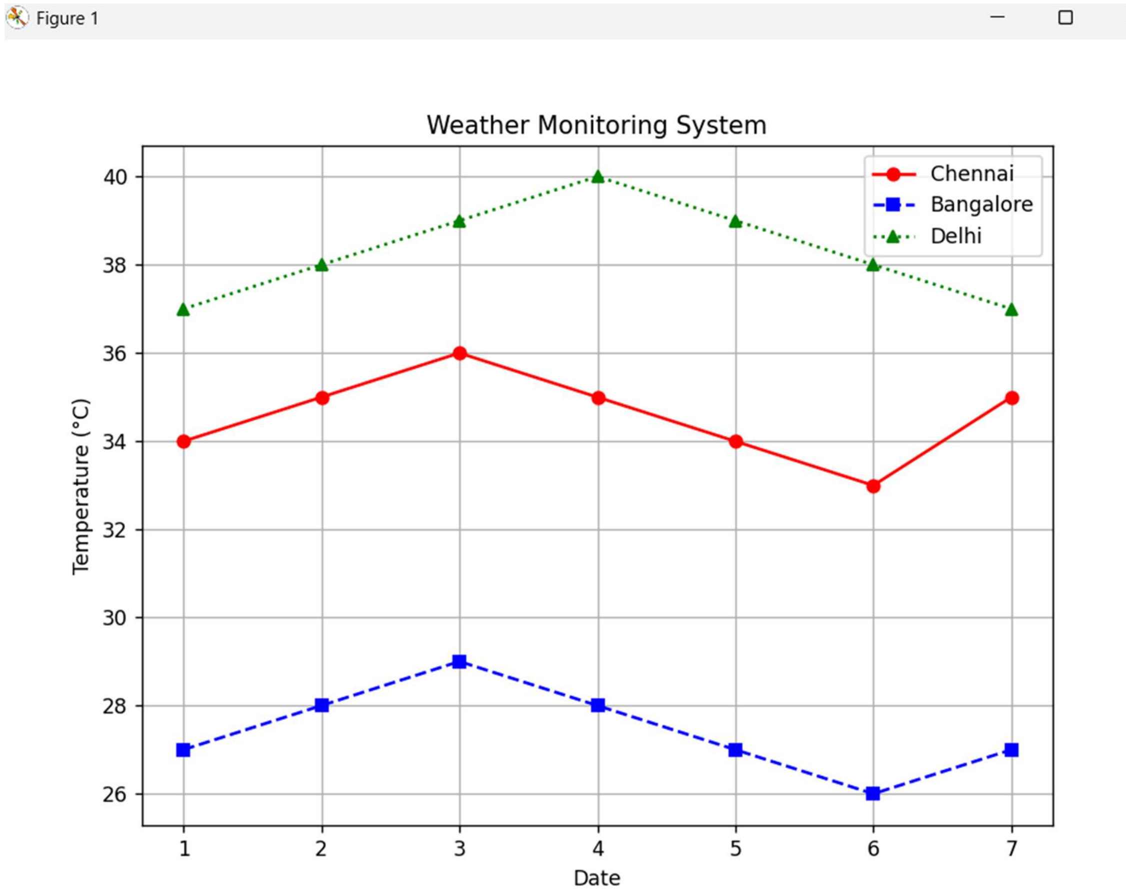


Scenario 4: Weather Monitoring System

```
import pandas as pd
import matplotlib.pyplot as plt

data={
"Date":[1,2,3,4,5,6,7],
"Chennai":[34,35,36,35,34,33,35],
"Bangalore":[27,28,29,28,27,26,27],
"Delhi":[37,38,39,40,39,38,37]
}

df=pd.DataFrame(data)
```

```
plt.figure(figsize=(8,6))

plt.plot(df["Date"],
         df["Chennai"],
         color="red",
         linestyle="-",
         marker="o",
         label="Chennai")

plt.plot(df["Date"],
         df["Bangalore"],
         color="blue",
         linestyle="--",
         marker="s",
         label="Bangalore")

plt.plot(df["Date"],
         df["Delhi"],
         color="green",
         linestyle=":",
         marker="^",
         label="Delhi")

plt.title("Weather Monitoring System")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
plt.legend()
plt.grid(True)
```

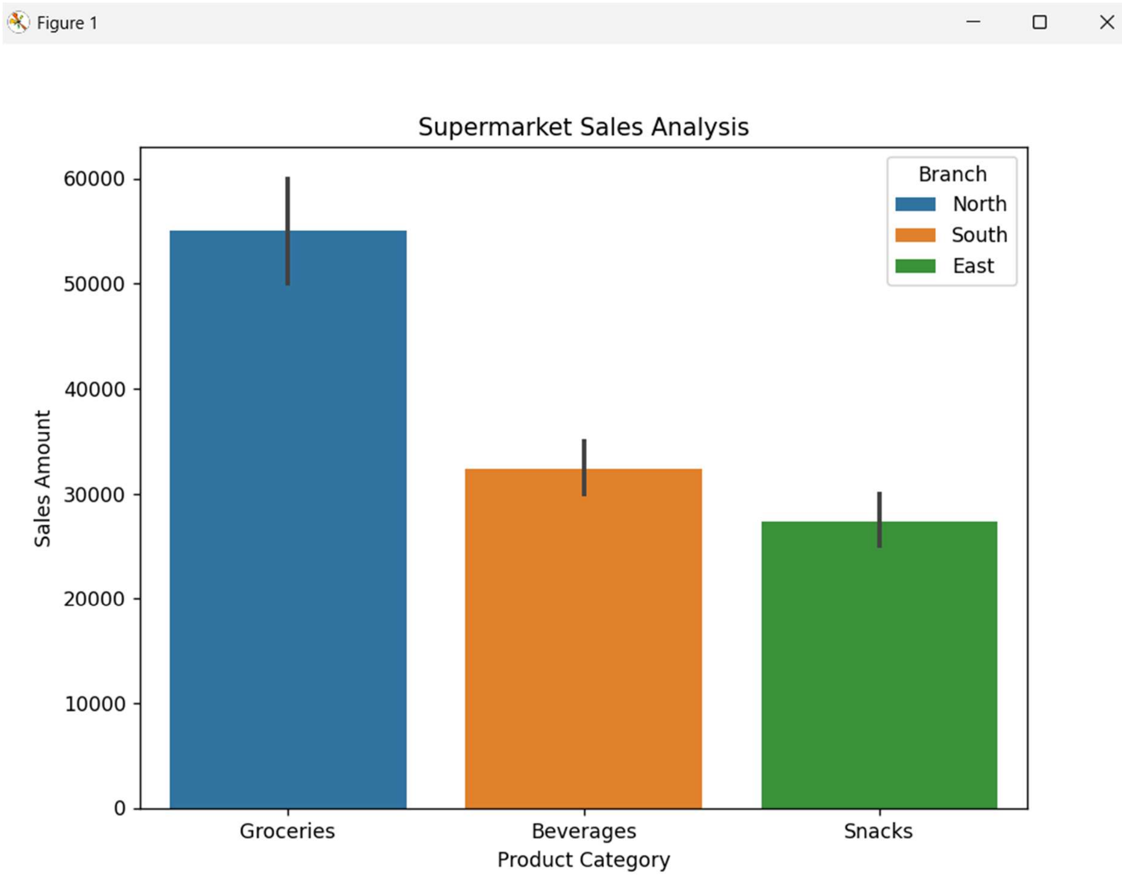
```
plt.show()
```



Scenario 5: Supermarket Sales Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data={
"Month":["January","January","January",
        "February","February","February",
        "March","March","March"],
"Product Category":["Groceries","Beverages","Snacks"]*3,
"Sales Amount":[50000,30000,25000,
             55000,32000,27000,
             60000,35000,30000],
```

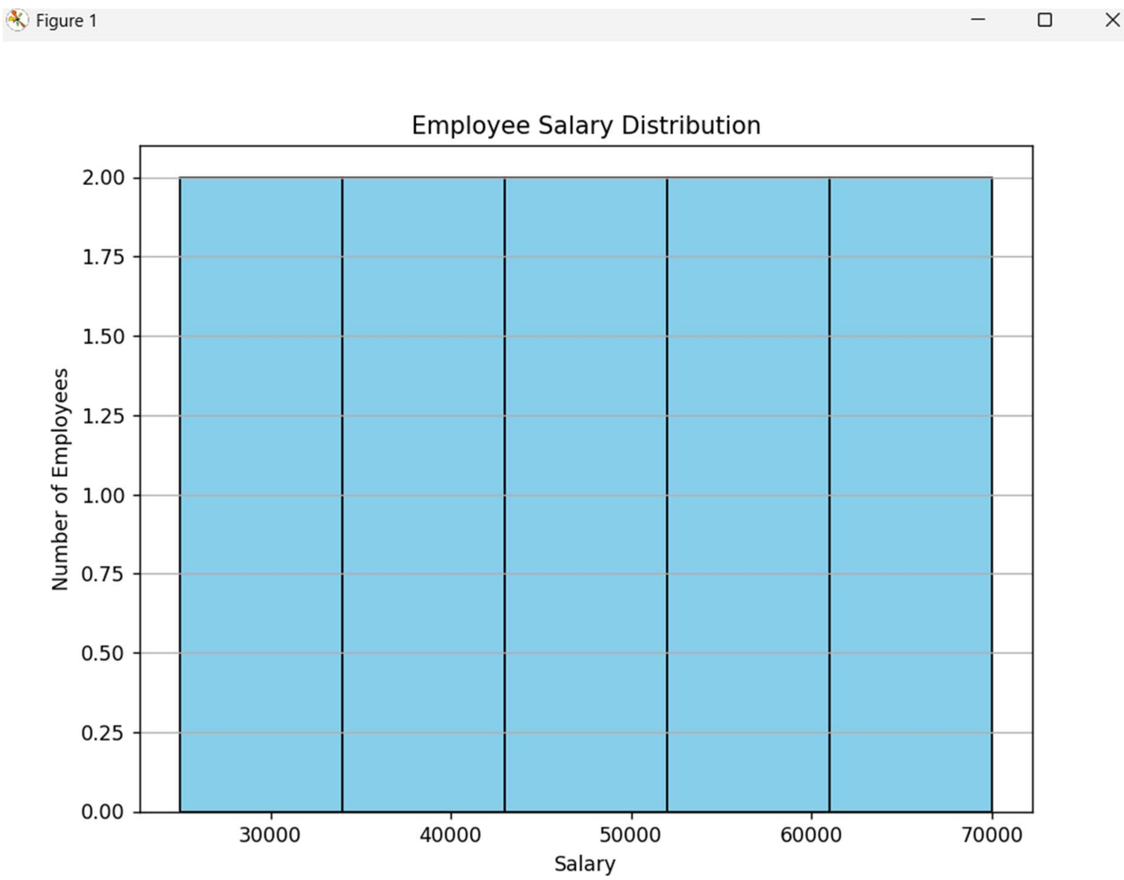
```
"Branch":["North","South","East",
        "North","South","East",
        "North","South","East"]
}

df=pd.DataFrame(data)

plt.figure(figsize=(8,6))

sns.barplot(data=df,
          x="Product Category",
          y="Sales Amount",
          hue="Branch")

plt.title("Supermarket Sales Analysis")
plt.xlabel("Product Category")
plt.ylabel("Sales Amount")
plt.legend(title="Branch")

plt.show()
```



Scenario 6: Employee Salary Distribution

```
import pandas as pd

import matplotlib.pyplot as plt


# Dataset

data = {

    "Employee ID":
["E001","E002","E003","E004","E005","E006","E007","E008","E009","E010"],

    "Salary": [25000,30000,35000,40000,45000,50000,55000,60000,65000,70000],

    "Experience": [1,2,3,4,5,6,7,8,9,10],

    "Department":
["HR","IT","Finance","Marketing","HR","IT","Finance","Marketing","HR","IT"]

}


df = pd.DataFrame(data)
```

```
plt.figure(figsize=(8,6))

plt.hist(df["Salary"],
         bins=5,
         color="skyblue",
         edgecolor="black")

plt.title("Employee Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Number of Employees")
plt.grid(axis="y")

plt.show()
```

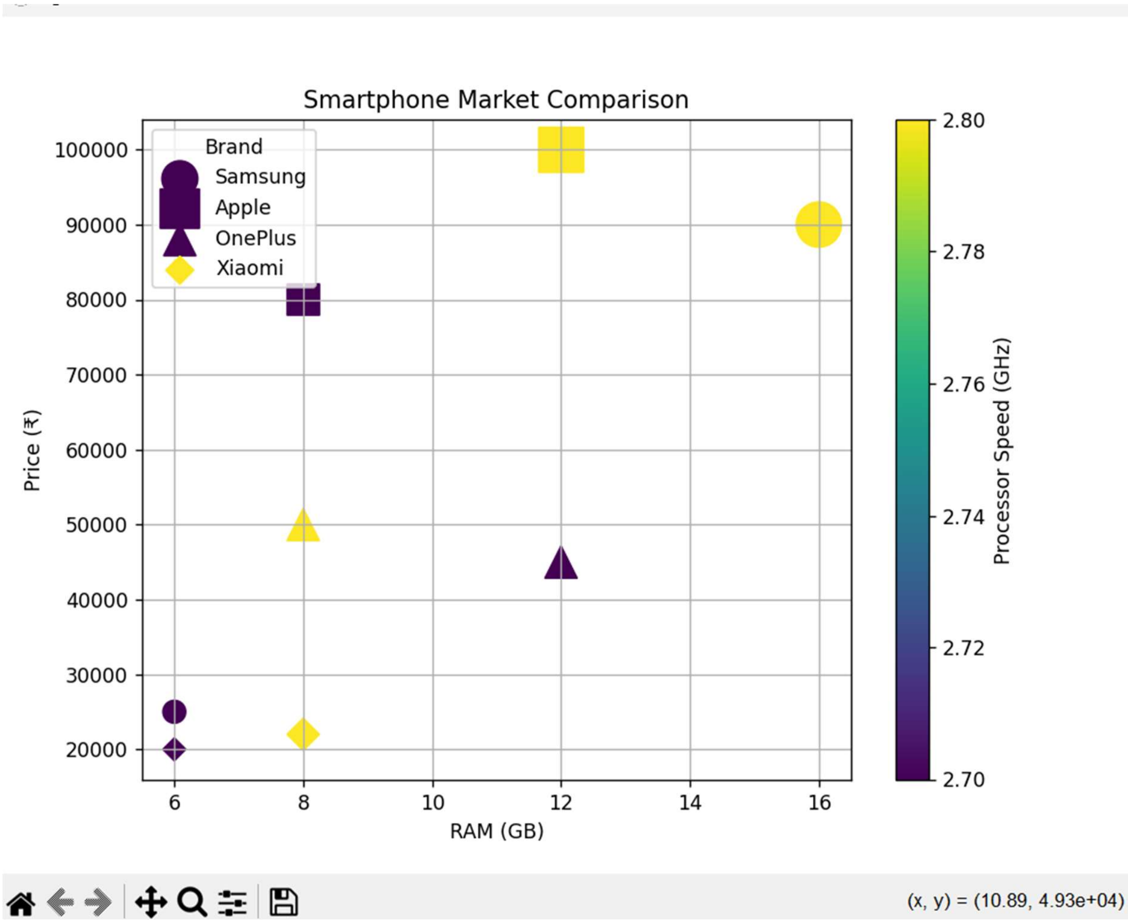


Scenario 7: Smartphone Market Comparison

```
import pandas as pd

import matplotlib.pyplot as plt


# Dataset

data = {

    "Brand":
["Samsung","Apple","OnePlus","Xiaomi","Samsung","Apple","OnePlus","Xiaomi"],

    "RAM": [6,8,12,8,16,12,8,6],

    "Processor Speed": [2.4,3.2,3.0,2.8,3.3,3.4,3.1,2.7],

    "Price": [25000,80000,45000,22000,90000,100000,50000,20000],

    "Storage": [128,256,256,128,512,512,256,64]

}


df = pd.DataFrame(data)


markers = {

    "Samsung":"o",

    "Apple":"s",

    "OnePlus":"^",

    "Xiaomi":"D"

}


plt.figure(figsize=(8,6))


for brand in df["Brand"].unique():

    temp = df[df["Brand"] == brand]


    plt.scatter(temp["RAM"],

                temp["Price"],

                s=temp["Storage"],
```

```
            c=temp["Processor Speed"],
            cmap="viridis",
            marker=markers[brand],
            label=brand)

plt.colorbar(label="Processor Speed (GHz)")
plt.title("Smartphone Market Comparison")
plt.xlabel("RAM (GB)")
plt.ylabel("Price (₹)")
plt.legend(title="Brand")
plt.grid(True)

plt.show()
import pandas as pd
import matplotlib.pyplot as plt

# Dataset
data = {
   "Brand":
["Samsung","Apple","OnePlus","Xiaomi","Samsung","Apple","OnePlus","Xiaomi"],
   "RAM": [6,8,12,8,16,12,8,6],
   "Processor Speed": [2.4,3.2,3.0,2.8,3.3,3.4,3.1,2.7],
   "Price": [25000,80000,45000,22000,90000,100000,50000,20000],
   "Storage": [128,256,256,128,512,512,256,64]
}

df = pd.DataFrame(data)

markers = {
   "Samsung":"o",
   "Apple":"s",
```

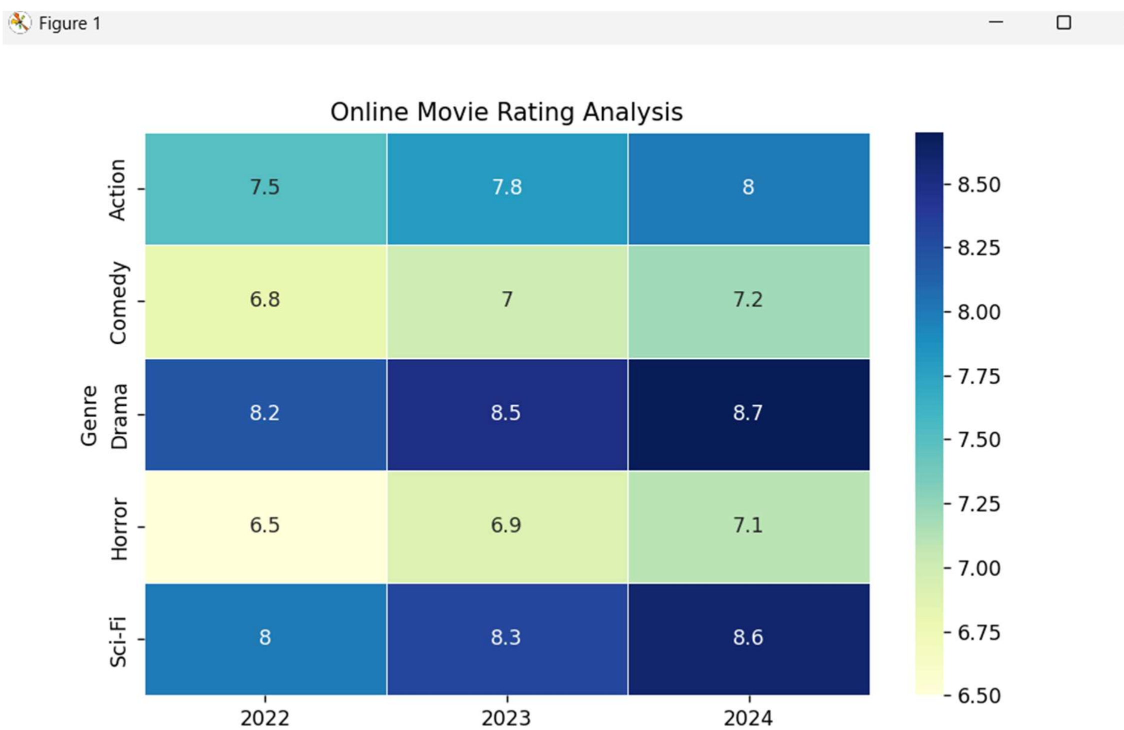
```
    "OnePlus":"^",
    "Xiaomi":"D"
}

plt.figure(figsize=(8,6))

for brand in df["Brand"].unique():
    temp = df[df["Brand"] == brand]

    plt.scatter(temp["RAM"],
                temp["Price"],
                s=temp["Storage"],
                c=temp["Processor Speed"],
                cmap="viridis",
                marker=markers[brand],
                label=brand)

plt.colorbar(label="Processor Speed (GHz)")
plt.title("Smartphone Market Comparison")
plt.xlabel("RAM (GB)")
plt.ylabel("Price (₹)")
plt.legend(title="Brand")
plt.grid(True)

plt.show()
```



scenario 8 online movie rating analysis

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


# Dataset

data = {

    "Genre": ["Action","Comedy","Drama","Horror","Sci-Fi"],

    "2022": [7.5,6.8,8.2,6.5,8.0],

    "2023": [7.8,7.0,8.5,6.9,8.3],

    "2024": [8.0,7.2,8.7,7.1,8.6]

}
```

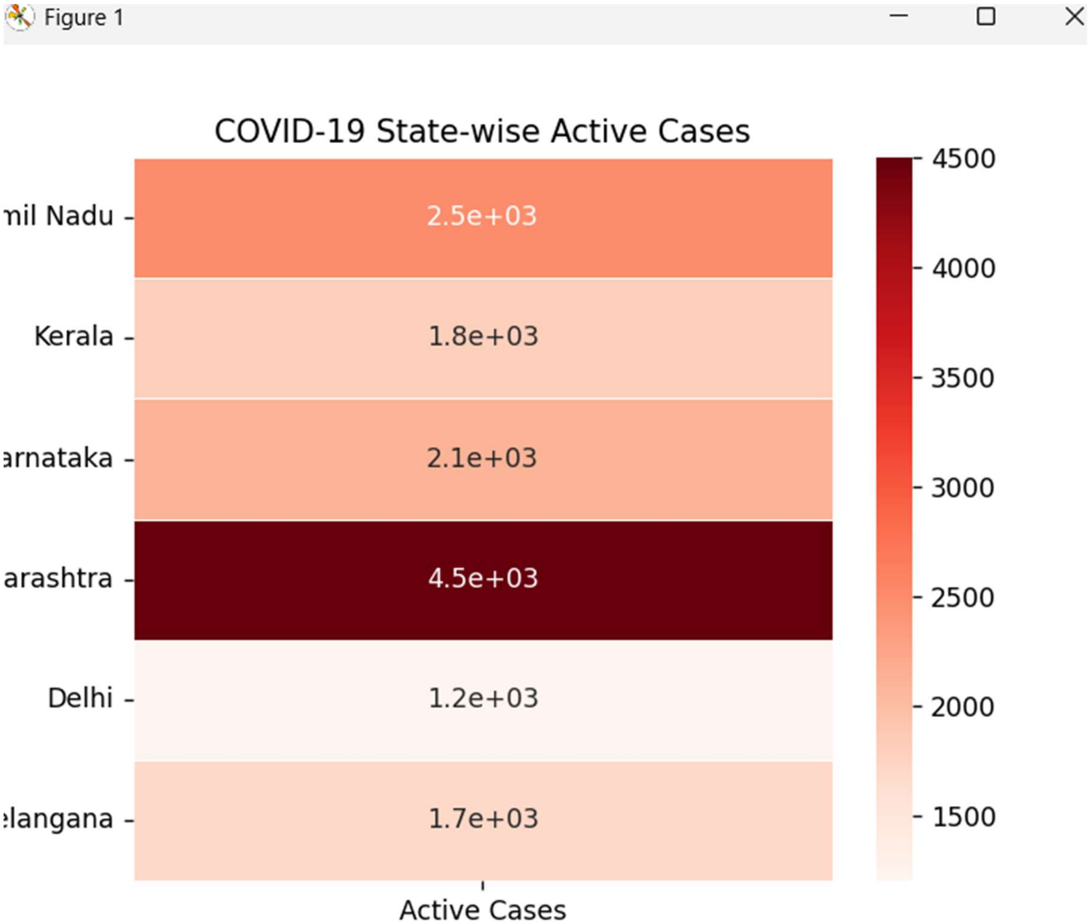
```
df = pd.DataFrame(data)

heatmap_data = df.set_index("Genre")

plt.figure(figsize=(8,5))

sns.heatmap(heatmap_data,
          annot=True,
          cmap="YlGnBu",
          linewidths=0.5)

plt.title("Online Movie Rating Analysis")

plt.show()
```



Scenario 9: COVID-19 State-wise Analysis (Tile Plot)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Dataset
data = {
    "State": ["Tamil Nadu","Kerala","Karnataka","Maharashtra","Delhi","Telangana"],
    "Active Cases": [2500,1800,2100,4500,1200,1700],
    "Vaccination Percentage": [92,95,91,89,94,90],
    "Population": [72000000,35000000,68000000,124000000,19000000,39000000]
}

df = pd.DataFrame(data)

tile_data = df.set_index("State")[["Active Cases"]]

plt.figure(figsize=(6,5))

sns.heatmap(tile_data,
            annot=True,
            cmap="Reds",
            linewidths=0.5)

plt.title("COVID-19 State-wise Active Cases")

plt.show()
```



Scenario 10: Iris Flower Classification

```
import pandas as pd

import matplotlib.pyplot as plt


# Dataset

data = {

    "Sepal Length":[5.1,4.9,5.8,6.0,6.5,6.3,5.0,6.7,6.9],

    "Sepal Width":[3.5,3.0,2.7,2.9,3.0,3.3,3.4,3.1,3.2],

    "Petal Length":[1.4,1.4,5.1,4.5,5.8,6.0,1.5,4.7,5.4],

    "Petal Width":[0.2,0.2,1.9,1.5,2.2,2.5,0.2,1.4,2.1],


"Species":["Setosa","Setosa","Virginica","Versicolor","Virginica","Virginica","Setosa","Versicolor","Virginica"]

}
```

```
df = pd.DataFrame(data)

colors = {
    "Setosa":"red",
    "Versicolor":"green",
    "Virginica":"blue"
}

markers = {
    "Setosa":"o",
    "Versicolor":"s",
    "Virginica":"^"
}

plt.figure(figsize=(8,6))

for species in df["Species"].unique():

    temp = df[df["Species"] == species]

    plt.scatter(temp["Sepal Length"],
                temp["Petal Length"],
                s=temp["Sepal Width"]*80,
                color=colors[species],
                marker=markers[species],
                label=species)

plt.title("Iris Flower Classification")
plt.xlabel("Sepal Length (cm)")
```

```
plt.ylabel("Petal Length (cm)")
plt.legend(title="Species")
plt.grid(True)

plt.show()
```